

MODULE 06 · DOMAIN 2 - DESIGN RESILIENT ARCHITECTURES

Durability, Backup and Disaster Recovery

Keeping the data, and getting it back.

DOMAIN 2 of 4 EXAM WEIGHT 26% MODULE 6 of 22

RTO is how long you can be down. **RPO** is how much data you can lose. Every DR question is decided by these two numbers.

The four disaster recovery strategies

STRATEGY	HOW IT WORKS	TYPICAL RTO AND RPO
Backup and restore	Back up data; build infrastructure when disaster strikes	Hours. Cheapest by far
Pilot light	Data replicated, core services off but ready to start	Tens of minutes
Warm standby	A scaled-down but running copy of everything	Minutes
Multi-site active-active	Full capacity running in both, serving live traffic	Near zero. Most expensive

ON THE EXAM

Cost rises and recovery time falls as you move down the table. Pick the cheapest row that meets the stated objective.

S3 durability and what it does not protect against

- ◆ **Eleven nines of durability** across at least three Availability Zones, in every storage class except One Zone-IA.
- ◆ **Durability is not backup.** A deleted or overwritten object is faithfully, durably gone.
- ◆ **Versioning** keeps previous versions, so deletes and overwrites become recoverable. Deletion adds a delete marker.
- ◆ **MFA Delete** requires a second factor to permanently delete a version. Protects against a compromised administrator.
- ◆ **Object Lock** makes objects immutable for a period. This is the answer for WORM and regulatory retention.

ON THE EXAM

'Protect against accidental deletion' means versioning. 'Cannot be deleted even by an administrator' means Object Lock.

RDS Multi-AZ or read replica?

MULTI-AZ

For availability

- Synchronous replication to a standby
- Automatic failover, same endpoint name
- Standby serves no traffic
- Different Availability Zone, same Region

READ REPLICA

For read scaling

- Asynchronous replication, so it can lag
- Manual promotion, and it gets a new endpoint
- Serves read queries, reducing primary load
- Can be in another Region

THE TIEBREAKER

The word 'reporting queries are slowing the database' means read replica. The word 'survive a zone failure' means Multi-AZ.

Replication and backup services

- ◆ **S3 Cross-Region Replication.** Asynchronous copy to another Region. Requires versioning on both buckets.
- ◆ **S3 Same-Region Replication.** For log aggregation or separating production and test data.
- ◆ **AWS Backup.** One policy covering EBS, RDS, DynamoDB, EFS, FSx and more, across accounts and Regions.
- ◆ **Aurora global database.** Cross-Region replication typically under a second, with fast promotion.
- ◆ **DynamoDB global tables.** Multi-Region, multi-active replication with no application changes.

ON THE EXAM

'Centralise and prove backup compliance across many services and accounts' is always AWS Backup.

Reading a DR question

01

Find the RTO

How long may it be down? Hours, minutes, or none.

02

Find the RPO

How much data may be lost? A day, minutes, or none.

03

Eliminate by RPO first

Backups every 24 hours cannot meet a 5-minute RPO. Cross it out.

04

Eliminate by RTO

Building infrastructure from scratch cannot meet a 10-minute RTO.

05

Take the cheapest survivor

Among options that meet both, cost decides.

ON THE EXAM

Most DR questions are solved by arithmetic, not by knowing services. Do the elimination in this order.

A retailer runs its order database on Amazon RDS. The business states that in a Region-wide failure it can tolerate being unavailable for up to four hours, but must not lose more than five minutes of orders.

Cost is a significant concern, and the finance director has questioned the proposed budget for a fully duplicated second environment.

Which disaster recovery strategy is MOST appropriate?

- ☐ A Backup and restore using nightly snapshots copied to a second Region
- ☐ B Pilot light with a cross-Region read replica and infrastructure defined as code
- ☐ C Warm standby running a scaled-down copy of the full environment
- ☐ D Multi-site active-active across two Regions

◆ PAUSE AND DECIDE

Which disaster recovery strategy is MOST appropriate?

A Nightly snapshots

Fails the RPO outright. A nightly snapshot can lose up to 24 hours of orders, not five minutes.

C Warm standby

Meets both objectives comfortably but runs a duplicate environment continuously, which costs more than the requirement justifies.

B Pilot light with cross-Region replica

Continuous replication keeps data loss to seconds, and four hours is ample to promote the replica and launch the rest.

D Multi-site active-active

The most expensive option, buying near-zero recovery time when four hours was acceptable.

THE SIGNAL

A five-minute RPO kills every snapshot-based option. A four-hour RTO allows the cheapest surviving strategy.

A company stores contracts in an S3 bucket. A regulator requires that once written, a contract cannot be modified or deleted by anyone, including the account root user, for seven years.

An internal audit also found that an engineer deleted a folder of drafts last year and it could not be recovered.

Which combination addresses both findings? (Choose two.)

- ☐ A Enable S3 Versioning on the bucket
- ☐ B Enable S3 Object Lock in compliance mode with a seven-year retention period
- ☐ C Enable server-side encryption with a customer managed KMS key
- ☐ D Apply a bucket policy denying the DeleteObject action

◆ PAUSE AND DECIDE

Which combination addresses both findings? (Choose two.)

- A** S3 Versioning
Makes accidental deletion recoverable, and it is also a prerequisite for Object Lock.
- B** Object Lock in compliance mode
In compliance mode no user, including root, can shorten the retention or delete the object. Exactly the regulatory requirement.
- C** Encryption with a KMS key
Protects confidentiality. It has no bearing on deletion or modification.
- D** Bucket policy denying deletes
A policy can be changed by anyone who can edit it, including root. It does not survive the 'including root' requirement.

THE SIGNAL

Governance mode allows privileged users to override. Compliance mode allows nobody. 'Including the root user' selects compliance mode.

An Amazon RDS for PostgreSQL database supports both a transactional application and a nightly analytics job. The analytics job runs long, complex queries that slow the application badly while it runs.

Separately, the business requires that a failure of the database's Availability Zone must not cause more than a brief interruption, and the application connection string should not need to change.

Which combination meets both requirements? (Choose two.)

- ☐ A Create a read replica and point the analytics job at it
- ☐ B Enable Multi-AZ deployment on the RDS instance
- ☐ C Increase the instance size to absorb the analytics load
- ☐ D Schedule the analytics job to run during a maintenance window

◆ PAUSE AND DECIDE

Which combination meets both requirements? (Choose two.)

- A** Read replica for analytics
Moves the heavy read workload off the primary. This is exactly what read replicas are for.
- B** Multi-AZ deployment
Automatic failover to a standby in another zone, and the endpoint name is unchanged, so the connection string stays the same.
- C** Larger instance
Buys headroom for the analytics problem at permanent cost, and does nothing at all for zone failure.
- D** Run during a maintenance window
Moves the pain rather than removing it, and still leaves the availability requirement unaddressed.

THE SIGNAL

This is the pair from the comparison slide, deliberately in one question. Read replica solves the read problem; Multi-AZ solves the availability problem.

A company runs workloads across sixty AWS accounts using EBS volumes, RDS databases, DynamoDB tables and EFS file systems. Each team currently manages its own snapshot scripts and schedules.

An auditor has asked the company to demonstrate that every production resource is backed up according to a single retention policy, and to produce evidence of compliance.

Which solution meets this with the LEAST operational overhead?

- | | |
|---|--|
| <p>A Standardise the existing scripts and run them from a central account</p> | <p>B Use AWS Backup with backup plans applied through AWS Organizations, and Backup Audit Manager for reporting</p> |
| <p>C Enable automated snapshots on each service and export a monthly inventory</p> | <p>D Use AWS Config rules to check that each resource has a recent snapshot</p> |

◆ PAUSE AND DECIDE

Which solution meets this with the LEAST operational overhead?

A Centralised scripts

Still bespoke code the company owns and maintains across sixty accounts. The opposite of least overhead.

C Per-service automated snapshots

Four different mechanisms with four retention settings, and a manual monthly report. No single policy, no automatic evidence.

B AWS Backup with Organizations

One policy covers all these services and every account, and Audit Manager produces the compliance evidence directly.

D Config rules checking snapshots

Detects whether backups happened but does not perform them. It reports on a problem it cannot fix.

THE SIGNAL

Many services plus many accounts plus a single policy plus evidence equals AWS Backup. The four together are its signature.

A global SaaS product must remain fully available if an entire Region fails, with no more than a few seconds of data loss and no manual intervention. Users are in Europe, North America and Asia, and latency complaints are frequent from Asia.

The application uses DynamoDB as its primary data store, and the budget explicitly allows for a multi-Region deployment.

Which approach best meets these requirements?

- ☐ A DynamoDB global tables with the application deployed in three Regions
- ☐ B DynamoDB with point-in-time recovery and nightly cross-Region backups
- ☐ C DynamoDB in one Region with DAX caches in the other two
- ☐ D DynamoDB with a read replica in each additional Region

◆ PAUSE AND DECIDE

Which approach best meets these requirements?

- A** Global tables in three Regions
Multi-active replication in seconds, local reads and writes for every user, and a Region failure leaves the others serving.
- B** PITR and nightly backups
Nightly backups cannot meet a few-seconds RPO, and restoring is manual intervention.
- C** DAX caches in other Regions
DAX is an in-Region cache for read latency. It holds no durable copy, so a Region failure loses the service.
- D** Read replicas per Region
DynamoDB does not have read replicas as a distinct feature. Cross-Region replication is provided by global tables.

THE SIGNAL

Global tables solve availability and latency in one feature. This is the rare case where multi-Region is genuinely correct.

A company's disaster recovery plan specifies a two-hour RTO. It relies on restoring EBS snapshots into a second Region and running CloudFormation templates to rebuild the environment.

The plan has never been tested. During a tabletop exercise, the team discovers the templates reference an AMI ID that exists only in the primary Region, and nobody knows how long a restore actually takes.

What should the solutions architect recommend FIRST?

- ☐ A Upgrade the strategy to warm standby to reduce reliance on the restore process
- ☐ B Run a full disaster recovery test to measure actual RTO and fix the errors it reveals
- ☐ C Copy all AMIs to the second Region and consider the plan complete
- ☐ D Increase snapshot frequency to improve the recovery point objective

◆ PAUSE AND DECIDE

What should the solutions architect recommend FIRST?

- A

 Upgrade to warm standby
Spends money to avoid a process nobody has measured. The current strategy may well meet the RTO once fixed.
- B

 Test it, measure it, fix it
An untested plan has an unknown RTO. Testing both validates the objective and surfaces the errors the exercise has already hinted at.
- C

 Copy AMIs and call it done
Fixes one known fault while leaving the plan untested. The next unknown fault surfaces during a real disaster.
- D

 More frequent snapshots
Improves RPO, which nobody said was a problem. The stated concern is recovery time.

THE SIGNAL

An untested DR plan does not have an RTO; it has an aspiration. Testing is the first recommendation, not an upgrade.

Module 6 takeaways

- ◆ **Find RTO and RPO first.** Eliminate on RPO, then RTO, then take the cheapest survivor.
- ◆ **Over-engineering is wrong.** Warm standby when pilot light meets the objective loses the mark.
- ◆ **Versioning** for accidental deletion. **Object Lock compliance mode** for 'not even root'.
- ◆ **Multi-AZ for availability, read replicas for read scaling.** Questions often need both.
- ◆ **Many services, many accounts, one policy, evidence** means AWS Backup.

ON THE EXAM

That completes domain two. Next we move to performance, where the question changes from 'does it survive' to 'is it fast enough'.